

AI-900 Practice Exam – Set 1 (60 Questions)



Microsoft Azure AI Fundamentals (AI-900)

This document contains a structured practice exam for **Microsoft Azure AI Fundamentals (AI-900)**. It is designed to help understand Azure AI services, Azure AI Foundry, machine learning concepts, and responsible AI principles.

Exam Overview

- Certification: Microsoft Certified: Azure AI Fundamentals (AI-900)
 - Level: Fundamental
 - Total Questions: 60
 - Format: Multiple Choice Questions (MCQ)
 - Key Focus Areas:
 - Azure AI Services
 - Azure AI Foundry
 - Generative AI and RAG
 - Azure Machine Learning basics
 - Content Safety and Responsible AI
 - Language, Vision, Speech, and Search services
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Instructions

- Each question has four options (A, B, C, D)
 - Select the most appropriate answer
 - Review explanations for understanding concepts
 - Focus on service selection and real-world scenarios
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Practice Questions – Set 1

1. RAG Hallucination Prevention

A healthcare provider is testing a RAG-based generative AI application in Azure AI Foundry. The AI must answer only from internal medical guidelines. What should they use to ensure responses are grounded?

- A. Prompt engineering only
- B. Azure AI Content Safety groundedness evaluation
- C. Azure Speech service
- D. Azure Translator

Answer: B

Explanation: Groundedness evaluation ensures that responses are based on retrieved documents and reduces hallucination risk.

2. Unified RAG Development Environment

Which Azure AI Foundry component allows visual orchestration, model comparison, and evaluation of LLMs?

- A. Azure Machine Learning Studio
- B. Prompt Flow in Azure AI Foundry
- C. Azure DevOps
- D. Azure Data Factory

Answer: B

Explanation: Prompt Flow enables workflow orchestration, evaluation, and model comparison.

3. Multi-Department AI Workspace

Which architecture supports centralized governance with isolated AI sandbox environments?

- A. Single subscription only
- B. Azure AI Foundry Hub and Project architecture
- C. One VM per team
- D. Azure Storage Accounts

Answer: B

Explanation: Hub-and-project structure provides centralized control with isolated environments.

4. Email Language Detection and Key Phrase Extraction

Which Azure service supports language detection and key phrase extraction?

- A. Azure AI Language
- B. Azure AI Vision
- C. Azure AI Speech
- D. Azure OpenAI

Answer: A

Explanation: Azure AI Language supports text analytics including language detection and key phrase extraction.

5. Real-Time Information Access for Agents

Which tool allows AI agents to access live external or web-based information?

- A. Code Interpreter
- B. Web grounding tool (Bing / external tool integration)
- C. Vector database
- D. Prompt Flow only

Answer: B

Explanation: Web grounding enables agents to fetch real-time information.

6. Content Safety Severity Levels

If Azure AI Content Safety is configured to block Medium and High severity, what happens to Low severity content?

- A. Blocked
- B. Allowed
- C. Modified automatically
- D. Deleted permanently

Answer: B

Explanation: Only Medium and High severity are blocked; Low severity is allowed.

7. Video, Audio, and Sentiment Analysis

Which Azure service provides transcription, face detection, and sentiment analysis in one solution?

- A. Azure AI Vision + Speech separately
- B. Azure AI Video Indexer
- C. Azure AI Translator
- D. Azure Form Recognizer

Answer: B

Explanation: Azure Video Indexer combines video intelligence, speech, and sentiment analysis.

8. Autonomous Agent Decision Making

Which Azure service enables agents to decide actions dynamically, including API calls and code execution?

- A. Azure Logic Apps
- B. Azure AI Foundry Agent Service
- C. Azure Data Factory
- D. Azure DevTest Labs

Answer: B

Explanation: Agent Service supports reasoning, tool use, and dynamic decision-making.

9. Prompt Injection via PDF

A malicious hidden instruction in a PDF uploaded to a RAG system is an example of:

- A. Data poisoning
- B. Prompt injection attack
- C. Model drift
- D. Overfitting

Answer: B

Explanation: Hidden instructions in external data sources are prompt injection attacks.

10. Evaluation Metric for Readability

Which metric evaluates grammar, structure, and readability of AI-generated output?

- A. Groundedness
- B. Coherence
- C. BLEU
- D. Recall

Answer: B

Explanation: Coherence measures fluency, readability, and logical flow.

11. Azure OpenAI Purpose

What does Azure OpenAI primarily provide?

- A. Database services
- B. Access to GPT models and generative AI capabilities
- C. Networking tools
- D. Storage services

Answer: B

Explanation: Azure OpenAI provides access to foundation models like GPT for generative AI applications.

12. Embeddings Used For

Embeddings are mainly used for:

- A. Audio processing
- B. Converting text into numerical vectors for similarity search
- C. Video rendering
- D. Image compression

Answer: B

Explanation: Embeddings convert text into vector representations for semantic similarity comparisons.

13. Vector Database Role

Why are vector databases used in AI systems?

- A. Store images only
- B. Store embeddings and enable fast similarity search

- C. Run virtual machines
- D. Train deep learning models

Answer: B

Explanation: Vector databases store embeddings and support semantic search in RAG systems.

14. Prompt Engineering

Prompt engineering refers to:

- A. Training GPUs
- B. Designing effective inputs to guide LLM outputs
- C. Building databases
- D. Network configuration

Answer: B

Explanation: Prompt engineering is the process of crafting inputs to improve model responses.

15. Temperature Parameter

What does increasing the temperature parameter in an LLM do?

- A. Makes output more deterministic
- B. Increases randomness and creativity
- C. Reduces token usage
- D. Prevents hallucination completely

Answer: B

Explanation: Higher temperature increases randomness in generated responses.

16. Azure AI Vision Use Case

Which task is supported by Azure AI Vision?

- A. Speech-to-text conversion
- B. Object detection in images
- C. Database indexing
- D. Model training automation

Answer: B

Explanation: Azure AI Vision is used for image analysis such as object detection and classification.

17. Text Analytics Capability

Azure AI Language supports:

- A. Video generation
- B. Sentiment analysis of text
- C. GPU provisioning
- D. Image rendering

Answer: B

Explanation: Azure AI Language provides sentiment analysis, key phrase extraction, and entity recognition.

18. AI Agent Tool Purpose

AI agents use tools primarily to:

- A. Reduce cloud cost
- B. Extend LLM capabilities with external systems and APIs
- C. Replace databases
- D. Eliminate prompts

Answer: B

Explanation: Tools allow agents to interact with APIs, data sources, and external systems.

19. Prompt Injection Attack

Prompt injection occurs when:

- A. Model training fails
- B. Malicious input manipulates model behavior or instructions
- C. Data is encrypted
- D. API is throttled

Answer: B

Explanation: Prompt injection tricks the model into ignoring system instructions.

20. Coherence Metric

The Coherence metric evaluates:

- A. Model speed
- B. Logical flow and readability of generated output
- C. Storage usage
- D. Network latency

Answer: B

Explanation: Coherence measures how logically consistent and readable the response is

21. Azure AI Search Role in RAG

What is the primary role of Azure AI Search in a RAG system?

- A. Model training
- B. Document retrieval using keyword and vector search
- C. Image generation
- D. Speech synthesis

Answer: B

Explanation: Azure AI Search retrieves relevant documents using hybrid (keyword + vector) search for RAG pipelines.

22. Machine Learning Classification Output

In a binary classification model, what does it predict?

- A. Continuous values
- B. Categories or labels
- C. Images
- D. Audio signals

Answer: B

Explanation: Classification models predict discrete labels such as Yes/No or Approved/Rejected.

23. Azure AI Language Use Case

Which task is supported by Azure AI Language service?

- A. Object detection
- B. Sentiment analysis

- C. Video rendering
- D. Model hosting

Answer: B

Explanation: Azure AI Language supports sentiment analysis, entity recognition, and key phrase extraction.

24. Supervised Learning Requirement

What is required for supervised learning?

- A. Unlabeled data
- B. Labeled training data
- C. Random data only
- D. No dataset required

Answer: B

Explanation: Supervised learning requires labeled input-output pairs.

25. Azure AI Content Safety Purpose

What is the main function of Azure AI Content Safety?

- A. Data storage
- B. Detect and filter harmful or unsafe content
- C. Train models
- D. Run virtual machines

Answer: B

Explanation: It detects harmful content such as hate speech, violence, and self-harm.

26. Embeddings in AI Systems

What is the purpose of embeddings?

- A. Store images
- B. Convert text into numerical vectors for semantic search
- C. Train networks
- D. Compress videos

Answer: B

Explanation: Embeddings represent text as vectors for similarity-based search.

27. RAG System Component

Which component retrieves external knowledge in a RAG system?

- A. LLM only
- B. Vector database or search index
- C. Speech service
- D. Vision API

Answer: B

Explanation: Retrieval systems like Azure AI Search provide relevant context to LLMs.

28. Azure OpenAI Use Case

What is Azure OpenAI primarily used for?

- A. Network monitoring
- B. Generating human-like text and content
- C. Database indexing
- D. Image compression

Answer: B

Explanation: Azure OpenAI provides access to GPT models for generative AI tasks.

29. Unsupervised Learning Example

Which scenario is unsupervised learning?

- A. Predicting loan approval
- B. Grouping customers based on behavior without labels
- C. Spam detection
- D. Image classification with labels

Answer: B

Explanation: Unsupervised learning finds patterns in unlabeled data.

30. (a)AI Agent Capability

What is a key capability of Azure AI Foundry agents?

- A. Only text generation
- B. Ability to use tools, APIs, and memory
- C. Only image processing
- D. Database hosting

Answer: B

Explanation: Agents can reason, call tools, and maintain memory for dynamic tasks.

30.(b) Model Context Protocol (MCP)

What is the primary purpose of Model Context Protocol (MCP) in Azure AI Foundry?

- A. Train machine learning models
- B. Standardize how agents access tools, memory, and external systems
- C. Store large datasets
- D. Replace Azure AI Search

Answer: B

Explanation: MCP provides a standard way for AI agents to interact with tools, APIs, and context sources.

31. Real-Time Speech Translation

Which Azure service enables real-time speech translation from one language to another?

- A. Azure AI Vision
- B. Azure AI Speech
- C. Azure AI Language
- D. Azure AI Search

Answer: B

Explanation: Azure AI Speech supports real-time speech-to-speech and speech-to-text translation.

32. Video Message Generation

Which Azure service can generate a video of a human avatar speaking a script?

- A. Azure AI Vision
- B. Azure AI Speech
- C. Azure AI Content Safety
- D. Azure AI Search

Answer: B

Explanation: Azure AI Speech (avatar/voice synthesis capabilities) enables realistic speech-based video generation.

33. Workflow Orchestration in AI Foundry

Which component is used to visually orchestrate multi-step AI workflows in Azure AI Foundry?

- A. Azure Data Factory
- B. Prompt Flow
- C. Azure Logic Apps
- D. Azure DevOps

Answer: B

Explanation: Prompt Flow is used to design, test, and deploy AI workflows.

34. Content Safety API Response Handling

When Azure AI Content Safety blocks a prompt, how is this communicated?

- A. HTTP 200 success
- B. HTTP 400/blocked response with safety category details
- C. Email notification
- D. Silent failure

Answer: B

Explanation: The API returns a structured response indicating the safety violation.

35. Temperature = 0 Effect

What happens when temperature is set to 0 in an LLM?

- A. Maximum creativity
- B. Completely random output
- C. Deterministic and highly consistent output
- D. Model stops responding

Answer: C

Explanation: Temperature 0 makes outputs deterministic and repeatable.

36. Vector Search Use Case

Which scenario best uses vector search?

- A. Sorting numbers
- B. Semantic search in documents
- C. Image rendering
- D. Network routing

Answer: B

Explanation: Vector search finds similar meaning-based content using embeddings.

37. Azure AI Document Intelligence Use

What is Azure AI Document Intelligence used for?

- A. Image generation
- B. Extracting structured data from documents
- C. Speech translation
- D. Video rendering

Answer: B

Explanation: It extracts text, tables, and fields from documents like invoices and forms.

38. Clustering in Machine Learning

Clustering is an example of:

- A. Supervised learning
- B. Unsupervised learning
- C. Reinforcement learning
- D. Deep reinforcement learning only

Answer: B

Explanation: Clustering groups data without labels.

39. Evaluation Metric: Groundedness

Groundedness in Azure AI Foundry evaluates:

- A. Model speed
- B. Whether responses are based on retrieved sources
- C. Network latency
- D. Storage size

Answer: B

Explanation: Groundedness ensures answers are supported by source documents.

40. AI Agent Memory Feature

What allows an AI agent to remember previous conversation context?

- A. Static prompts
- B. Agent memory / session context
- C. Batch processing
- D. Logging only

Answer: B

Explanation: Agent memory maintains conversation history across interactions.

Question 41

A healthcare provider is testing a RAG-based generative AI application in Azure AI Foundry. The AI is designed to answer patient questions based exclusively on internal medical guidelines. To ensure the AI is not hallucinating answers from the public internet, which feature should be used?

Options:

- A. Groundedness Evaluation
- B. Temperature Control
- C. Token Limit Configuration
- D. Batch Inference

Correct Answer: A

Explanation:

Groundedness evaluation ensures the model's response is based strictly on provided source data and not external knowledge.

Question 42

Which Azure AI Foundry capability allows you to compare multiple LLMs like GPT-4o and Mistral using the same prompts and evaluate performance before deployment?

Options:

- A. Prompt Flow Evaluation
- B. Azure Data Factory
- C. Azure Logic Apps
- D. Azure DevOps Pipelines

Correct Answer: A

Explanation:

Prompt Flow Evaluation enables testing multiple models with consistent prompts and scoring metrics like relevance and groundedness.

Question 43

What is the best Azure AI Foundry architecture for managing centralized AI governance while allowing isolated development environments?

Options:

- A. Single Resource Group Model
- B. Multi-Tenant Foundry Hub and Project Structure
- C. Standalone VM Deployment
- D. Local Docker Environment

Correct Answer: B

Explanation:

Foundry Hub-and-Project architecture provides centralized control with isolated project-level environments.

Question 44

Which Azure service can detect language, extract key phrases, and process email text in one unified service?

Options:

- A. Azure AI Language
- B. Azure Blob Storage
- C. Azure Functions
- D. Azure Monitor

Correct Answer: A

Explanation:

Azure AI Language supports language detection, key phrase extraction, and entity recognition.

Question 45

An AI agent needs real-time access to weather and news. Which tool should be added in Azure AI Foundry Agent Service?

Options:

- A. Code Interpreter
- B. Bing Grounding Tool
- C. SQL Connector
- D. Vector Index Tool

Correct Answer: B

Explanation:

Bing Grounding Tool allows agents to retrieve real-time external information.

Question 46

If Azure AI Content Safety is set to block Medium and High severity violence, what happens to Low severity content?

Options:

- A. It is blocked
- B. It is allowed
- C. It is escalated
- D. It triggers retry

Correct Answer: B

Explanation:

Only Medium and High are blocked; Low severity is allowed.

Question 47

Which Azure service can perform speech transcription, face detection, and sentiment analysis in videos?

Options:

- A. Azure AI Video Indexer
- B. Azure AI Search
- C. Azure Data Lake
- D. Azure Synapse Analytics

Correct Answer: A

Explanation:

Video Indexer supports multi-modal video analysis including speech, faces, and sentiment.

Question 48

Which capability enables an AI agent to dynamically decide whether to call APIs or run Python code?

Options:

- A. Static Prompting
- B. Autonomous Agent Reasoning
- C. Batch Processing
- D. Manual Script Execution

Correct Answer: B

Explanation:

Autonomous agents can reason, plan, and execute tools dynamically.

Question 49

How do you block a competitor name like “DataVault” in Azure AI Content Safety?

Options:

- A. Fine-tuning
- B. Custom Blocklist
- C. Model retraining
- D. Prompt Engineering

Correct Answer: B

Explanation:

Custom blocklists allow explicit term filtering.

Question 50

What type of attack involves hidden instructions inside a PDF that manipulate AI behavior?

Options:

- A. Data Poisoning
- B. Prompt Injection Attack
- C. Model Drift
- D. Overfitting

Correct Answer: B

Explanation:

Prompt injection embeds malicious instructions to override model behavior

Question 51

Which metric in Azure AI Foundry Evaluation Flow measures grammatical correctness and readability of generated text?

Options:

- A. Groundedness

- B. Coherence
- C. Recall
- D. Similarity

Correct Answer: B

Explanation:

Coherence evaluates whether the response is logically structured, grammatically correct, and easy to understand.

Question 52

Which Azure service is best for extracting text from handwritten scanned historical documents?

Options:

- A. Azure AI Vision
- B. Azure AI Document Intelligence
- C. Azure AI Search
- D. Azure Bot Service

Correct Answer: B

Explanation:

Document Intelligence (formerly Form Recognizer) is designed for OCR and handwritten document extraction.

Question 53

How is private network-only access enforced in Azure AI Foundry for highly secure environments?

Options:

- A. Public endpoint restriction only
- B. VNet integration and Private Link
- C. API key rotation
- D. Token expiration policy

Correct Answer: B

Explanation:

Private Link and Virtual Network integration ensure no public internet exposure.

Question 54

Which Azure AI Content Safety feature protects against prompt injection attacks?

Options:

- A. Groundedness Check
- B. Prompt Shields
- C. Content Filters
- D. Rate Limiting

Correct Answer: B

Explanation:

Prompt Shields detect and block malicious prompt injection attempts.

Question 55

Where can you compare benchmark performance (MMLU, HumanEval) of LLMs in Azure AI Foundry?

Options:

- A. Model Catalog
- B. Azure Monitor
- C. Azure Portal Metrics
- D. Azure DevOps

Correct Answer: A

Explanation:

Model Catalog provides benchmark comparisons of different foundation models.

Question 56

Which tool allows no-code machine learning model training and automated algorithm selection?

Options:

- A. Azure Machine Learning AutoML
- B. Azure Data Factory
- C. Azure Logic Apps
- D. Azure CLI

Correct Answer: A

Explanation:

AutoML automatically selects best-performing ML models without coding.

Question 57

Which Azure ML tool provides drag-and-drop visual pipeline design?

Options:

- A. Azure ML Designer
- B. Azure Data Studio
- C. Azure Synapse Pipeline
- D. Azure Fabric Notebooks

Correct Answer: A

Explanation:

Azure ML Designer provides a visual canvas for ML workflows.

Question 58

Which ML technique is used when there are no predefined labels for clustering customer tickets?

Options:

- A. Supervised Learning
- B. Reinforcement Learning
- C. Unsupervised Learning
- D. Transfer Learning

Correct Answer: C

Explanation:

Unsupervised learning is used for clustering without labels.

Question 59

Which evaluation metric measures False Negatives in classification models?

Options:

- A. Precision
- B. Recall
- C. Accuracy
- D. F1 Score

Correct Answer: B

Explanation:

Recall measures how many actual positives were correctly identified; it reflects False Negatives.

Question 60

What is the effect of setting temperature to 0.1 in Azure OpenAI models?

Options:

- A. More creative responses
- B. Highly random outputs
- C. More deterministic and focused responses
- D. Increases hallucinations

Correct Answer: C

Explanation:

Low temperature makes responses deterministic, stable, and less random.